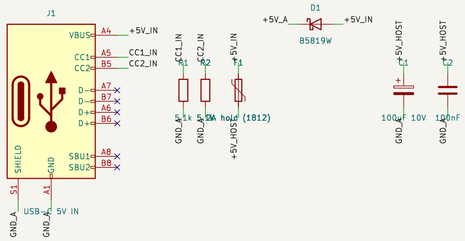
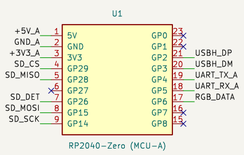
Power input — 5 V adapter on USB-C (power only)

Adapter 5 V, 2–3 A. F1 protects the USB-A port. D1 feeds MCU-A so the keyboard works while PC 1 is off.
Note: with PC 1 switched off and its cable plugged in, ~4.7 V reaches PC 1 VBUS through the module. Variant B avoids this.

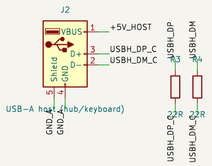


MCU-A — USB host, router, microSD (connects to PC 1 through its own USB-C)

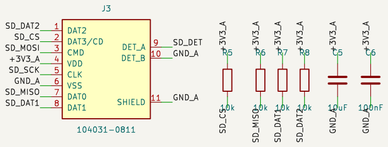
GP2/GP3 = PIO-USB host D+/D−. GP4/GP5 = UART1 TX/RX. GP14/15/28/29 = SPI1 SCK/MOSI/MISO + CS. GP26 = card detect. GP16 = on-board WS2812.



USB host port (keyboard, mouse, dongle or hub)

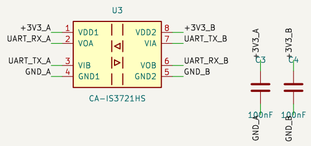


microSD — shared drive

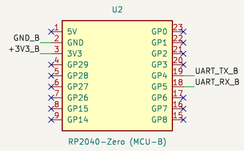


Galvanic isolation between PC 1 and PC 2 (GND_A and GND_B never meet)

CA-IS3721HS (pin compatible with ADuM1201). UART runs up to ~3 Mbaud: check the grade you buy.

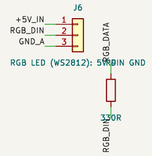


MCU-B — connects to PC 2 through its own USB-C



Optional RGB LED header (WS2812/SK6812): 5V, DIN, GND

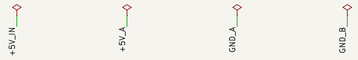
Data from GP6 through R9. Most WS2812B/SK6812 accept 3.3 V data at 5 V supply; keep the wire short.



Mechanical



Power flags



kmswitch (open hardware)

Sheet: /
File: kmswitch-a.kicad_sch

Title: kmswitch — variant A (RP2040-Zero carrier)

Size: A2 | Date: | Rev: 1.0
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | Id: 1/1